AQ4.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

Find the rank of the following matrix: $\begin{bmatrix} 0131 \\ 1001 \\ 3203 \end{bmatrix}$ $\begin{bmatrix} \\ 013102300113 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Consider the following set of vectors SS in R^3 R3 to answer the questions 2, 3 and 4.
S = S = {(1, 2, 0), (0, 3, 1), (3, 3, − 1), (3, 0, − 2)(1, 2, 0), (0, 3, 1), (3, 3, −1), (3, 0, −2)}.
1 point

Let VV be the vector space spanned by the vectors (1, 2, 0) and (0, 3, 1)(1, 2, 0) and (0, 3, 1). Which one of the following is correct?

- ○ V = {$(a, 2a + 3b, b) \mid a, b \in R$}V = {(a, 2a + 3b, b) | a, b ∈ R}
- ○ V = {$(0, 2a - 3b, b) \mid a, b \in R$}V = {(0, 2a − 3b, b) | a, b ∈ R}
- ○ V = {$(-a, 2a + 3b, 0) \mid a, b \in R$}V = {(−a, 2a + 3b, 0) | a, b ∈ R}
- ○ V = {$(0, 2a + 3b, 0) \mid a, b \in R$}V = {(0, 2a + 3b, 0) | a, b ∈ R}

No, the answer is incorrect.
Score: 0
Accepted Answers:
V = {$(a, 2a + 3b, b) \mid a, b \in R$}V = {(a, 2a + 3b, b) | a, b ∈ R}

If the vectors in SS are written as the columns of a matrix AA, then what will be the rank of AA?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

What will the dimension of the vector space spanned by SS be?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

1 point

Choose the set of correct options from the following.

- [ ] Row rank and column rank of a matrix is always the same.
- [ ] The rank of a zero matrix is always 0.
- [ ] The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.
- [ ]

Rank of the $n \times n$n × n identity matrix is 1 for any $n \in N$n ∈ N.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Row rank and column rank of a matrix is always the same.
The rank of a zero matrix is always 0.
The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.

Level 2

1 point

Consider the following upper triangular matrix to choose the correct options.

$$\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$ $\begin{bmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{bmatrix}$

where $a, b, c, d, e, f \in R$a, b, c, d, e, f ∈ R.

- [ ] If $f = 0$f = 0, then the rank of the matrix must be less than or equal to 2.
- [ ] If $f = 0$f = 0, then the rank of the matrix must be exactly 2.
- [ ] If a, b, c, d, e, fa, b, c, d, e, f are all non-zero then the rank of the matrix must be 3.
- [ ] If a, d, fa, d, f are non-zero then the rank of the matrix must be 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $f = 0$f = 0, then the rank of the matrix must be less than or equal to 2.
If a, b, c, d, e, fa, b, c, d, e, f are all non-zero then the rank of the matrix must be 3.
If a, d, fa, d, f are non-zero then the rank of the matrix must be 3.

If rank of the matrix $\begin{bmatrix} 2 & -3 & 4 \\ 0 & 1 & -2 \\ 1 & -3 & a \end{bmatrix}$ $\begin{bmatrix} 2 & 0 & 1 \\ -3 & 1 & -3 \\ 4 & -2 & a \end{bmatrix}$ is 2 then find the value of aa.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

Find the rank of the matrix AA, where $A = [a_{ij}]$A = [aij] is of order $3 \times 3$3 × 3 and $a_{i,j} = \min\{i, j\}$ ai,j = min{i, j}, $i, j = 1, 2, 3$i, j = 1, 2, 3.

[Hint: Write down the matrix and apply row operations to derive it's reduced row echelon form.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

What is the rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{bmatrix}$A = [1xx21yy21zz2], where x, y, zx, y, z are distinct non-zero integers?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Rank of a $3 \times 3$3 × 3 matrix AA is 3 and another $3 \times 3$3 × 3 matrix BB is 2. What is the minimum possible value for the rank of ABAB?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Check Answers

Your score is: 0/10